

function calls and database joins. This shift required restructuring data flow, implementing service discovery, and addressing challenges such as data consistency, API versioning, and centralised logging. Despite the complexity, the transition significantly improved scalability, maintainability, and the ability to deploy updates without affecting unrelated parts of the system.

In the process of converting the job portal from a monolithic to a microservices architecture shown in Figure 3, several mathematical and computational models were utilised to ensure optimal system performance and efficiency. Queuing theory, analyse request handling and inter-service communication were applied to the model, helping to minimise latency and avoid bottlenecks. Load balancing algorithms, such as round-robin and least connections, were evaluated to distribute traffic evenly across services. Dependency graphs were used to identify tightly coupled components in the monolith, guiding the service decomposition strategy. Additionally,

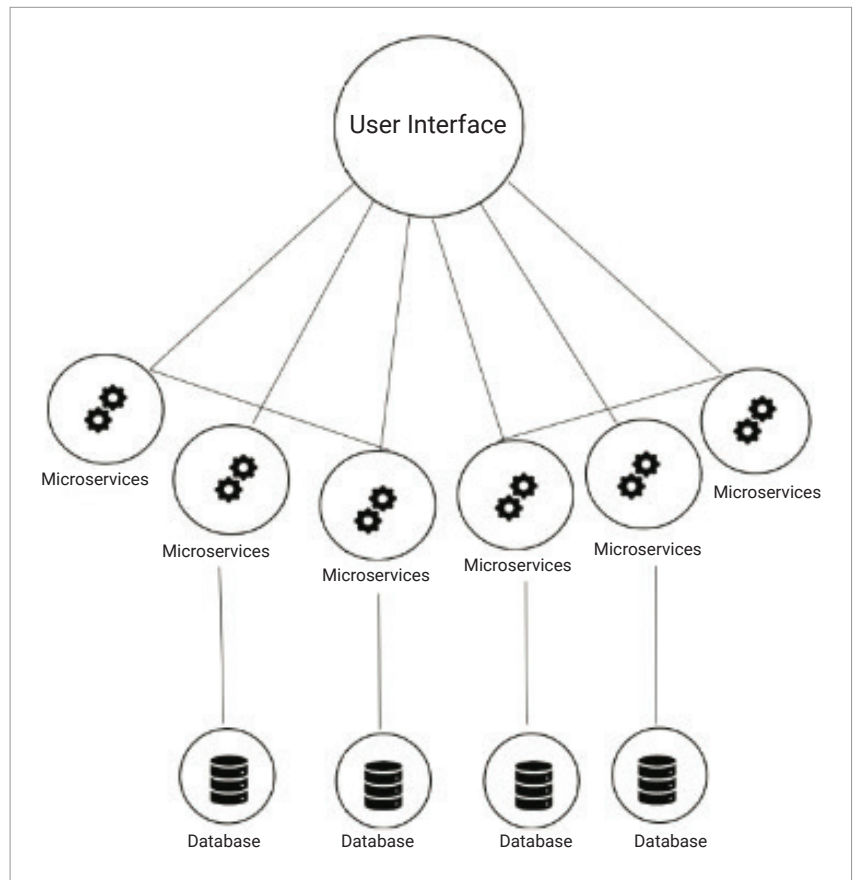


Figure 2: Microservices architecture

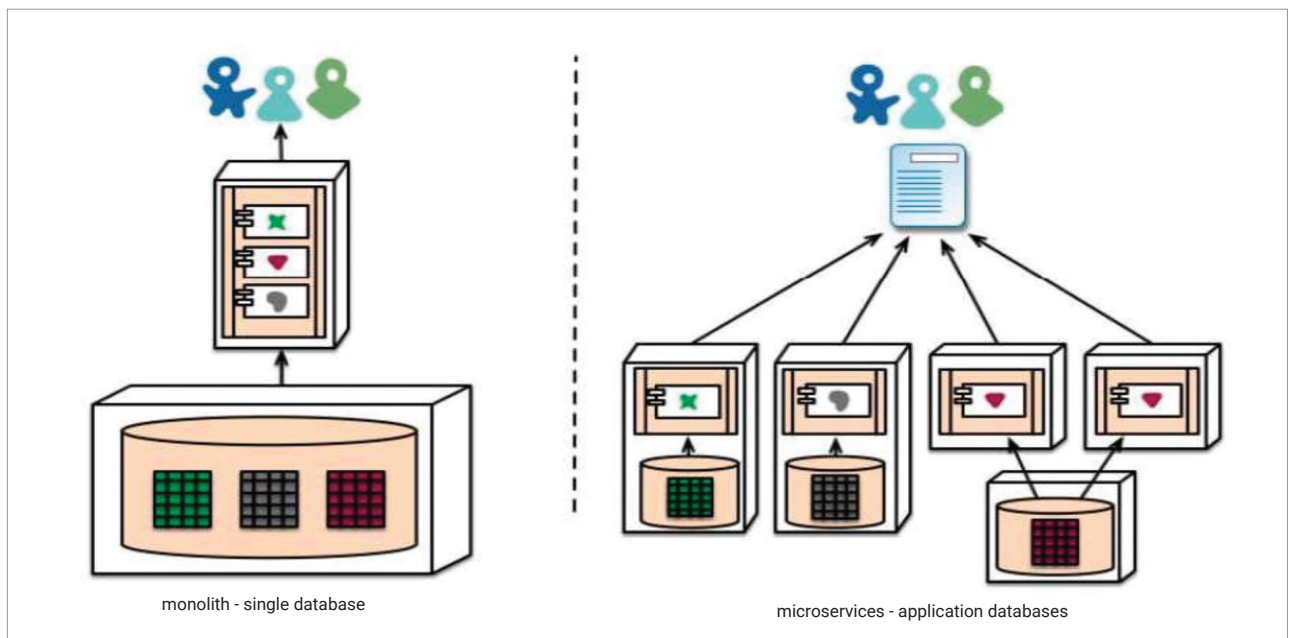


Figure 3: Transition process overview